Changing Human percept of an ambiguous moving plaid to unambiguous by the noise and the direction of Random Dot Kinematograms

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1. Introduction

One of the best useful tools for psychophysicists to investigate visual motion detection is Random Dot Kinematograms (RDKs). Over the last few decades, random dot kinematograms have emerged as an effective psychophysical stimulus to evaluate low-level motion processing.

Random dot kinematograms where the percentage of dots that move together within the same direction, as if attached to a flat, rigid surface, is varied while the remaining dots move randomly across trials, is that the foremost commonly used measurement of motion coherence thresholds. It requires global integration of motion across several points. The motion coherence thresholds are determined by measuring the percentage of coherently moving dots required for accurate detection of coherent motion or direction discrimination. Each dot has a limited lifetime so the detection and interpretation of coherent motion requires a global integration of multiple motion signals. Participants report with a button press whether the motion is in one direction or another (left or right; up or down) or which sub-regions contain coherent motion. The perceived coherence is directly related to activity in area MT, but to not activity in earlier visual areas.

The ability to understand the direction of a moving object within the surroundings is an important visual performance. however, how does the brain estimate the direction of a moving object in clutter? In primates, directional transient responses square measure thought to be created in V1 by local motion detectors, which may be implemented in many ways that (Derrington et al., 2004); namely correlation-type motion detector (Reichardt, 1961), null direction inhibition model(Barlow & Levick, 1965), and motion energy filter model (Adelson & Bergen, 1985). In the next stage, directional short-range filters (Braddick, 1974) give rise to directional V1 simple cells by accumulating directional transients over relatively short spatial and temporal ranges.

Cortical motion analysis starts with direction-selective neurons within the primary cortical area. local speed measurements can differ depending on the contour orientation of the moving object and the receptive field size and the spatial position of the motion detector. due to their small receptive fields, V1 neurons have a limited view of a moving object and inscribe only the part of motion perpendicular to a contour, which might be different from the true motion of the entire object. At the following stage of visual motion analysis, area MT within the superior temporal sulcus, that has larger receptive fields, could be a good candidate site for motion

integration (Lisberger & Movshon, 1999). Pursuit initiation is tightly linked with the activity and properties of direction-selective cells in area MT (Lisberger & Movshon, 1999) and lesions in MT impair smooth pursuit initiation for stimuli presented in the corresponding visual field (Newsome et al., 1985). The control of pursuit during the steady-state pursuit, when the moving target is stabilized on the retina, cannot be accounted for by MT neurons, because they are silent in the absence of retinal motion (Newsome et al., 1988).

The question of how the human visual system pools motion signals across totally different speeds, directions, and spatial positions have received some attention. This pooling method appears to be quite flexible. for example, it's been shown that the balance between spatial segmentation and integration are often adapted to satisfy the task necessities for perception (Burr et al., 2009) and that form cues can contribute to that selection process (Maruya et al., 2010). Contextual motion changes the perceived direction of bistable motion (Baker & Graf, 2010). Additionally, quite a few solutions to resolve global motion seem to exist in parallel and compete against each other(Bowns & Alais, 2006). Interestingly, mechanism-based decoding algorithms provide a better explanation of perception than simple stimulus-based statistics (Webb et al., 2007).

There are various types of RDKs i.e. Transparent motion, White noise, Brownian motion, etc. the definition of these three types is as follow:

1. **Transparent motion:** In this condition, signal and noise dots formed two distinct populations. Dot lifetime was 200 ms for both signal and noise dots. Signal dots moved in the signal direction and were randomly repositioned at the end of their lifetime. Noise dots were assigned a random direction and kept this direction until the end of their lifetime. When the lifetime of a noise dot ended, it was repositioned and assigned a new random motion direction. We call this stimulus transparent motion because subjects usually perceive it as two superimposed transparent surfaces (Snowden & Verstraten, 1999).
2. **White Noise (WN):** A new set of signal dots are randomly chosen to move in the signal directly from each frame to the next, and the remaining (noise) dots are randomly relocated; i.e., each noise dot is given random direction and speed (Britten et al., 1992, 1996).
3. **Brownian Motion (BM):** This is similar to WN, but all dots move with the same speed; i.e., noise dots are only given random directions (Seitz et al., 2005; Seitz & Watanabe, 2003).

Random-dot displays can contain varied strengths of motion signal and mix signal and noise within the same part of visual space. as a result of these advantages, random-dot stimuli are usually used in psychophysical and neuroscience studies.

Using a stimulus as a visual distractor in our settings, we can only place our stimulus in particular parts of the visual fields, either central or in the outer periphery. Using a random dot kinematogram, we would like to investigate what the thresholds are on aperture and coherence such that the coherent direction influences the subject's gaze in at least half of the trials. Also, for the aperture, we want to use a bistable moving paradigm. It is one among the mechanisms controlling visual perceptual bistability once two perceptual solutions are obtainable – by controlling the duration of every perception. Moving plaids are an example of such ambiguity once two moving gratings are superimposed, they can be perceived as having either transparent motion (the gratings slide across each other) or coherent (pattern) motion. In general, the dominance of one or other percept depends on intrinsic properties of the plaid, such as speed, orientation, and spatial frequency of the element gratings (Hupé & Rubin, 2003) and the global orientation of the plaid (Hupé & Rubin, 2004).

1.1. Hypothesis

Based on the results of our literature study we conclude that the only algorithm that has an optimal performance in response to a different parameter is Brownian motion. If putting Brownian motion random dots on the gratings of the moving plaid and by changing the noise and direction as a factor and analyzing the subject’s perception of the direction of ambiguous moving plaids we might change the subject’s percept and guide them by random dot kinematograms.

If we could inference this hypothesis, we may also put another factor to experiment. Furthermore, if my hypothesis is correct the next question for me is where would be the context of using this hypothesis?

One group of subjects could be the Autism Spectrum Disorder (ASD) children. ASD children have higher motion coherence thresholds than typical children that means ASD children need a higher level of coherent moving dots to detect a coherent direction in movement. Subject with ASD demonstrated less perceptual sensitivity in the direction discrimination task compared to controls(Kaiser & Shiffrar, 2009).

Through our study, we analyze several works on RDKs and moving plaids and also the literature using a combination of both. There is various literature study the influence of different parameter on subject’s perception and their eye movement based on types of random dot kinematograms.

Late acceleration, pursuit gain, and perceived speed all relied on kinematogram type, with the sensible agreement between pursuit gain and perceived speed (Schütz et al., 2010). in this study, they investigate how the human visual system and the pursuit

system react to visual motion noise. they used three types of RDKs and 5 levels of coherence. For transparent motion, pursuit gain and perceived speed were independent of coherence level. For white and Brownian motions, pursuit gain and perceived speed increased with coherence but were higher for white than for Brownian motion. Their study suggests that the pursuit system based on their condition and different form of noise integrates across all directions of motion but not across all speeds.

They used different types of noise in their study from the study of (Scase et al., 1996) that described three motion algorithms:

1. **random position:** each noise dot is given a random direction and displacement
2. **random walk:** each noise dot is given a random direction, but the same displacement as signal dots
3. **random direction:** similar to random walk but each dot moves only in its designated direction, which is randomly assigned at the beginning, whenever it is chosen to be noise

under same and different conditions like in (Williams & Sekuler, 1984), with nominal variations in dot density and speed.

They also study the ability of the smooth pursuit system to integrate signals and discard noise. From the smooth pursuit system, we go through the study of anticipatory eye movements.

Anticipatory smooth eye movement studied the expectations of motion of random-dot kinematograms (RDKs) where the signals of the future direction of global RDK motion are cued either before the onset of the RDK or before a change in its direction of motion (Santos et al., 2012).

Anticipatory smooth eye movements can be caused in response to expectations about the direction of global motion. However, cues about other properties of the global motion (e.g., dot lifetime) shown to be ineffective. According to this view, anticipatory smooth eye movements should be considered in the same way as other perceptual decisions (Yang et al., 2010).

Study on psychophysical comparison of random dot algorithm shown that parametric and algorithmic differences can produce dramatically different performances that may be concluded from the pertinent neurophysiological data regarding spatiotemporal displacement tuning properties of cells in area MT and how the tuning operates changes with input contrast and retinal eccentricity (Pilly &

Seitz, 2009). from the results of this study, we managed to conclude that the only prediction that turned out to be correct was an optimal performance in response to algorithm Brownian motion. the relative performances for other algorithms were shown to depend on parameters such as contrast, aperture size, spatial displacement, and temporal displacement, but not speed.

There are a few studies associated with both random dot kinematograms and bistable moving plaids and one of the approach studies through this subject is about analysis of the contribution of extrinsic factors to perception; specifically contextual motion and eye movements(Baker & Graf, 2010).

In this paper, extrinsic factors that affect plaid perception and can be extended, studied. surrounding motion can influence perception when it is matched in speed and direction to one of the bistable percepts, mostly by increasing percept durations in the surrounding direction.

For moving stimuli, it is conceivable that surrounds ‘capture’ eye movements, producing involuntary saccades in the surround direction it perceived from this study that saccades were not related with perceptual transitions, though blink rate increased around the time of a switch. This specifies that saccades do not cause switches, yet saccades in a congruent direction might help to prolong a percept.

2. Project Protocol

Our first step was about studying the literature and study the effect of RDKs and moving plaids on the visual percept. Through this step, we manage to figure out how the further steps of this project would be and could manage to design an aperture to test the initial hypothesis.

Based on our hypothesis the other steps to initiate this experiment would be designing the stimuli and the aperture based on figure 2 that was extracted from the study on the neural mechanism of perceptual hysteresis in the human visual system (Sayal et al., 2020; Sousa et al., 2018) on Python. A very useful toolbox for this experiment is the Psychopy toolbox that is written both for Python and Matlab programming. Most other studies have written their code on Matlab and so design and developing these stimuli on Python would be a very innovative way. In the next section, the experimental protocol of this project is expressed in detail.

Also, at the end of the designing the experiment, gathering the required elements such as necessary documents and finding a suitable subject for implementing this

experiment is needed. The final step of this project is to analyze the obtained data and conclude whether the initial hypothesis was correct or not.

2.1. Experimental Paradigm

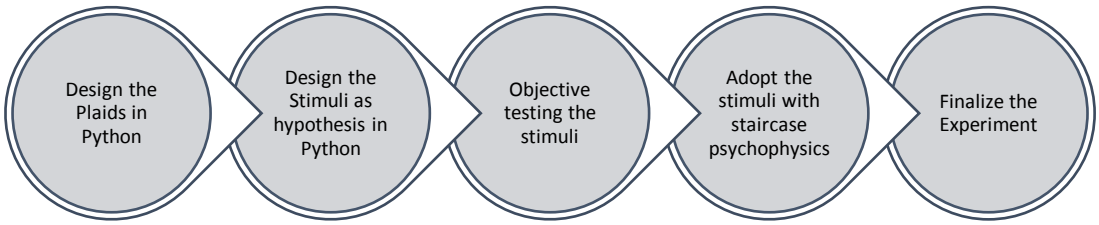

Figure 1 – Experimental Protocol summary chart

2.1.1. Python Coding – Design the Plaids

In the beginning, I wanted to design a stimulus as in figure 2. This figure was simulated in Matlab for a neurological experiment (Sousa et al., 2018). This stimulus represents a plaid that can be interpreted as two gratings moving horizontally in two opposite directions (Transparent) or one pattern that goes upward (Coherent). And also, RDKs are moving only on the gratings. The first step of my experimental design was how to just represent moving plaids in Python.

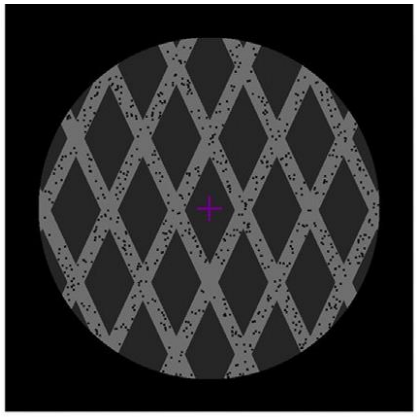

Figure 2 - Moving plaid with Random dot kinematograms

2.1.2. Python Coding – Hypothesis Presentation

Based on my initial hypothesis, I wanted to direct the perception of the subject by manipulating the RDKs direction. So, I had to design the whole stimuli with the plaids and dots on them and the velocity of dots and plaids had to be compatible.

2.1.3. Objective Testing the Stimuli

After successfully designing the stimuli, the process began to objective testing in regards to the initial hypothesis. As a person so adopted to the experiment, I wasn't

a good subject because I knew in which direction the plaids were supposed to move. So, with the great help of my supervisors, I managed to troubleshoot my experiment.

2.1.4. Adoptive Staircase Procedure

Staircase: A psychophysical method introduced in 1962 by the US cognitive scientist Tom Norman Cornsweet (born 1929) in which, for absolute thresholds, a variable stimulus is presented repeatedly and is adjusted upwards whenever it is not perceived and downwards whenever it is perceived, and for difference thresholds a variable stimulus is adjusted to increase its absolute difference from a standard stimulus whenever the difference is not discriminated or is adjusted to decrease its absolute difference from the standard stimulus whenever the difference is discriminated. Also called the up-and-down method. Compare method of average error, method of constant stimuli, method of limits.

The simplicity and flexibility of adaptive staircases have led to their adoption as the procedures of choice in many laboratories. These methods generally use the previous one or more responses within an adaptive track to select the next trial placement, then provide a threshold estimate in a variety of ways, most commonly by averaging the levels at the direction reversals in the adaptive track (i.e., the turnaround points). Simple up-down staircases call for a reduction in stimulus level when the subject's response is positive (e.g., "I hear a tone") and an increase in stimulus level when the response is negative. Figure 3 shows an example of a simple up-down adaptive track. Beginning at a level above threshold, positive responses lead to continued decreases in stimulus level until a negative response occurs. This triggers a reversal in the direction of the track, and levels on subsequent trials increase until the next change in response. The simple up-down staircase procedure targets the 50% performance level on a psychometric function that extends from 0% correct performance at the chance to 100% correct performance. In other words, the track targets the stimulus level for which the probability of a correct response equals the probability of an incorrect response or, equivalently, the level at which the track would move up or down on the stimulus axis with equal probability. The value of this type of procedure is in the very few assumptions necessary for its implementation. The only necessary assumption for use of these methods is a monotonic relationship between stimulus levels and performance levels (Li et al., 2013).

As for this experiment, I needed at least 80 percent of performances so the transformed staircase had been used and as I had three different percepts (left, right,

up), I had to use the interleaved staircase, one for coherent stimuli and one for transparent stimulus.

2.1.5. Finalize the experiment

The last part of the experiment was choosing a place with keeping some variables constant (e.g. luminosity, height and distance of a subject to a screen, size of the aperture) and I also structured a small questionnaire (Appendix A) to collect the subject's condition at a particular time of the experiment.

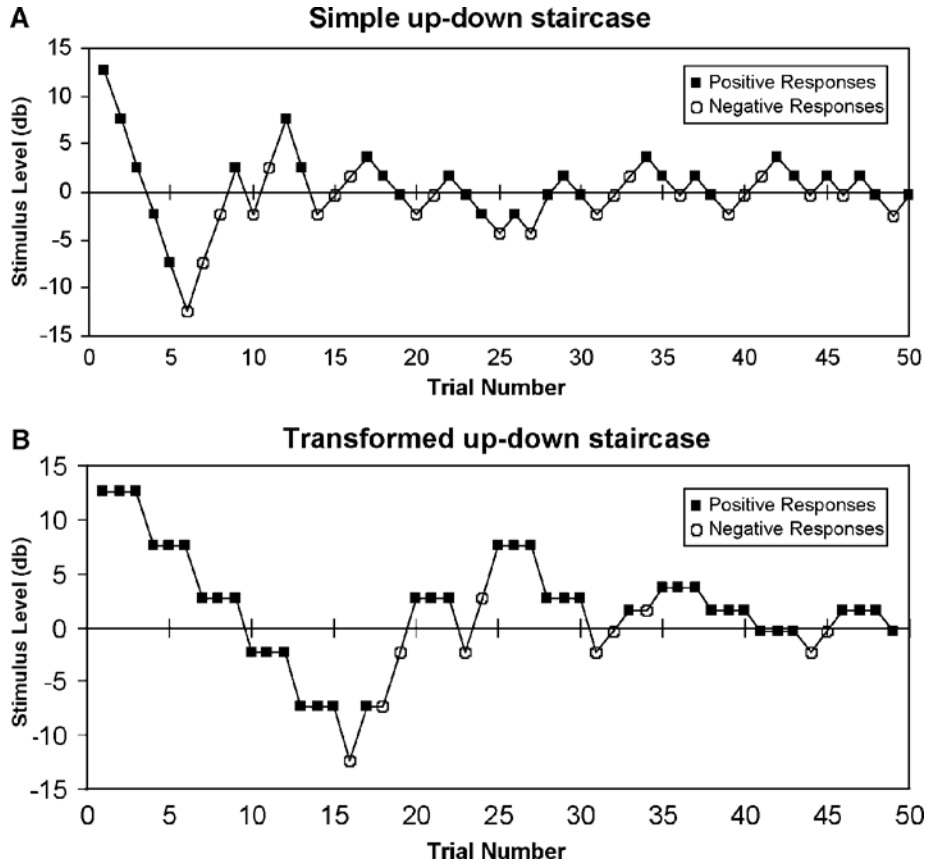

Figure 3 - Adaptive tracks following a staircase procedure. (A) Simple up-down staircase; (B) transformed up-down staircase, following a three-down, one-up algorithm.

3. Methods

3.1. toolbox

The python toolbox that used for this project was psychopy. This library is written and developed by the University of Nottingham.

PsychoPy is an open-source package for running experiments in Python (a real and free alternative to Matlab). PsychoPy combines the graphical strengths of OpenGL with the easy Python syntax to give scientists a free and simple stimulus presentation

and control package. It is used by many labs worldwide for psychophysics, cognitive neuroscience, and experimental psychology.

Some most used modules from this library were GratingStim, DotStim from the visual class of this library.

3.2. Methods Encounters

The first challenge of the process of designing an ambiguous plaid with psychopy was the fact that psychopy generates and modifies a drifting plaid similar to figure 4.

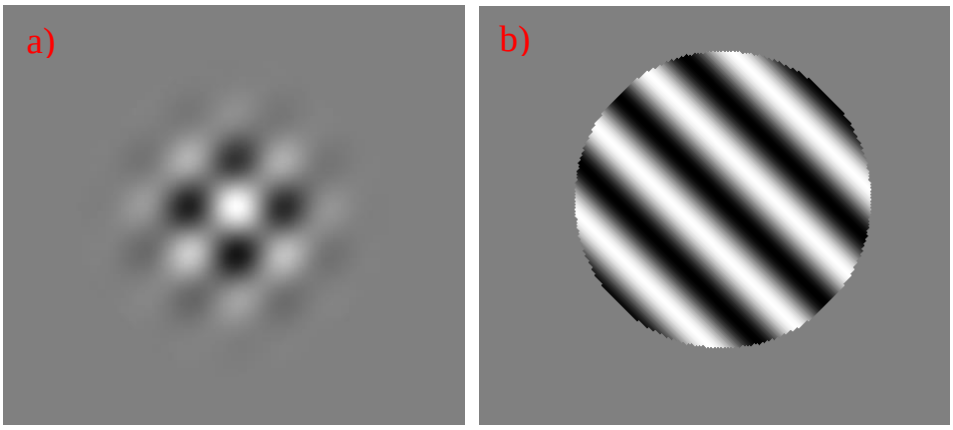

Figure 4 – a) psychopy view of a plaid, b) Modified using GratingStim variables

This stim was far from the plaids we needed and the furthest we could obtain just by GratingStim variables. Also, there were not many tools to modify these plaids. The main idea of generating the required moving plaids was initiated by Professor Phlypo from the work of Ms. Juliette LENOUVEL on a similar plaid.

The idea was to consider two gratings (left and right) and build the gratings from scratch instead of modifying predefined models. in this initiation, the place of the grating's pixel is defined and assigned a value between 0 to 1 and assigned 1 to the rest of the plaids.

and then by measuring the mean transparency of this aperture and normalize the alpha of the whole aperture. figure 5 shown the idea behind the design

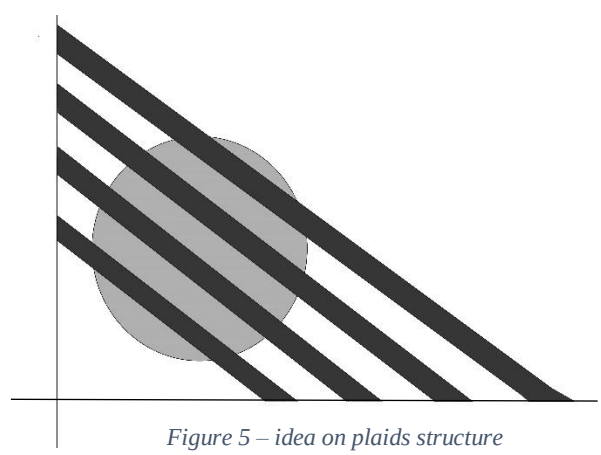

Figure 5 – idea on plaids structure

of the plaids. The acquired plaids are shown in figure 6.

The next contest in here before reaching the process of implementing dots on the plaids is how to move these gratings.

After defining the pixels on the aperture as the plaids, the whole acquired texture is used in the GratingStim to produce the patterns of gratings. Now we have a set of gratings that its angle can change during the time of the trial and this change can be possible by varying the phase variable of the stim and by considering the aperture as a circle that contains these gratings it seems that the phase of the circle is changing and it looks like the plaids are moving.

Figure 6 – The final plaids obtained by defining a texture from scratch

The next challenge in this project was how to put the random dots on the whole aperture but only visible on the gratings and not visible on the space in the patterns between the gratings.

Several methods taken into account and implemented:

The first method was to define a large number of different small sets of dots varied in the position and puts together for every grating that was moving together with the same speed. There were several problems in this method, first, the set of each dot was too small to be visible and to have an effect. Second, the sets move together to the end of the window instead of the aperture and they wouldn't return to the start point. and the third, based on the fact that we factorized the level of the noise in the RDKs, by this method, the set was implemented couldn't be able to reduce its number of dots per plaid and it wasn't applicable. This method performed poorly

and I interpret from this method that the best method for the project to be applicable is to only define one set of dots and make it visible only on the plaids.

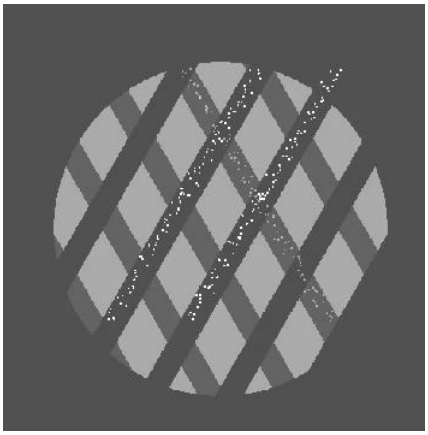

Figure 7- First method of creating the stimulus

The second method that considered was to contemplate the aperture in three levels:

- The main plaids
- The dots
- The inverse of the plaids

The inverse of the plaids considered as when the alpha of the space between the patterns changes to the alpha of the gratings and on the other hand, turning the grating's alpha to one.

However, this method didn't give us the right answer but during this trial, an important feature of Psychopy showed itself.

In Psychopy when adding two plaids together, the alpha of the middle space reaches the maximum and doesn't allow anything in the middle space to be visible. This implemented method is shown in figure 8.

So, instead of defining three-level, it can be applied in defining the aperture with two levels:

- The dots
- The main gratings with a double in the texture

After overcoming the appearance of dots on the plaids the next subject that was important for this experiment was the velocity of the dots and the plaids. we implement the plaids to move by changing its phase and it is based on radian per second, the dots velocity is based on degree per second. Based on the initial hypothesis, and some trial and errors we conclude that it's best if the speed of the

dots would be compatible with the speed of the gratings. Based on this conclusion velocity of the plaids converted by the equation (1) and (2) to become the speed of the dots and become the same speed either in radian or degree per second in the x-direction and the y-direction.

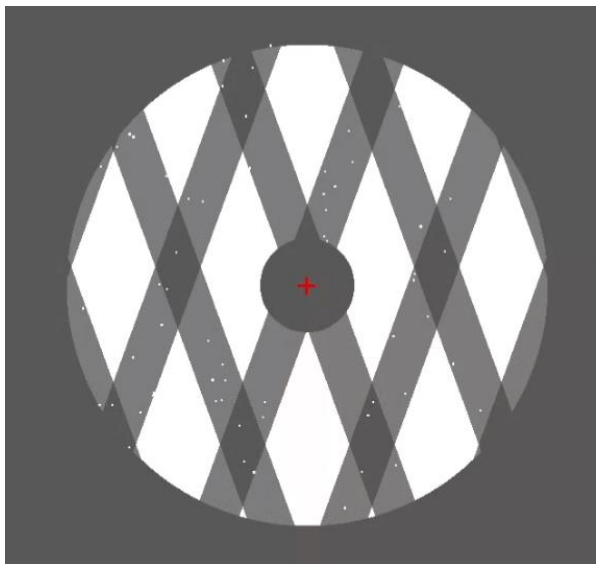

Figure 8 – obtained the result on putting dots on the gratings by doubling the texture

$$Velocity_{dots_x} = \tan^{-1}\left(\frac{\tan(v)}{\cos(orient)}\right) * \frac{180}{\pi} \quad (1)$$

$$Velocity_{dots_y} = \tan^{-1}\left(\frac{\tan(v)}{\sin(orient)}\right) * \frac{180}{\pi} \quad (2)$$

v is the velocity of the plaids.

After some objective tests and since the plaid is supposed to be ambiguous and the pattern where two gratings over passing by each other may affect by the change in the orientation of the plaids, this method improved and the final velocity produced by the current spatial frequency of the plaid and the orientation.

Angular Velocity:

$$V(x,y,z) = \begin{cases} \dfrac{y}{\cos\left(\frac{\pi}{180}(x-z)\right)} & ,\cos\left(\frac{\pi}{180}(x-z)\right) > \frac{1}{2} \\ y * \dfrac{tan\left(\frac{\pi}{180}(x-z)\right)}{\sin\left(\frac{\pi}{180}(x-z)\right)} & ,\cos\left(\frac{\pi}{180}(x-z)\right) \leq \frac{1}{2} \end{cases} \quad (3)$$

Angular Frequency:

$$F(x,y,z) = \begin{cases} \left| y * \cos\left(\frac{\pi}{180}(x-z)\right) \right| & , \cos\left(\frac{\pi}{180}(x-z)\right) > \frac{1}{2} \\ \left| y * \frac{sin\left(\frac{\pi}{180}(x-z)\right)}{\tan\left(\frac{\pi}{180}(x-z)\right)} \right| & , \cos\left(\frac{\pi}{180}(x-z)\right) \leq \frac{1}{2} \end{cases} \quad (4)$$

$$sf_x = F(0,0.2,20) \quad (5)$$

$$sf_y = F(90,0.2,20) \quad (6)$$

$$V_y = V(90,2.5,20) \quad (7)$$

$$Velocity_x = V_y * sf_x * \cos\left(\frac{\pi}{180}(v)\right) \quad (8)$$

$$Velocity_y = V_y * sf_y * \sin\left(\frac{\pi}{180}(v)\right) \quad (9)$$

v is the orientation of the plaids.

$$Velocity_{dots_x} = \min\big(V(x, 2.5,20)\big) \quad (10)$$

$$Velocity_{dots_y} = \min\big(V(x, 2.5, -20)\big) \quad (11)$$

x is the direction of the dots in degree.

The main hypothesis of this project was that changing the direction of the dots on the ambiguous plaid, it may cause the subject to perceive the direction of the plaids in unambiguous form, and two factors were stated.

1) direction
2) noise (to prove the effects of dots number on the experiment)

As for this project, three levels of noise are considered 10, 50, 90. Also, 0 is the baseline at the start of the experiment.

Other levels of noise were discussed. but by increasing the number of the level, the time of the trial would be increased without having any effect and also the other level that was rejected was [25%,50%,75%]. but since the noise was considered as a factor and 100 percent coherence would be neglected the noise, the complete dot set without noise was set aside. and by the large gap between 100 and 75 percent that might cause some effect of the noise on the data won't be considered, the final level was decided on [10%,50%,90%] dot coherency.

3.3. Designing the Experiment

At the start of the project, the experiment was supposed to be as in the hypothesis just the dots moving across the gratings and get the percepts of the participant on the direction. After some trial and discussion, we decided to add some other trials to the experiment. This trial was as follow:

1- unambiguous plaids without the dots and changing the alpha of the gratings: for instance, when the direction is supposed to be left, the alpha of the left gratings becomes darker and the alpha of the right gratings becomes lighter and counter wise for the right direction. As for the upward direction, both alphas would be equal and the plaids would ambiguous.
2- plaids with dots without changing the alpha (ambiguous plaid).
3- plaids with dots and changing the alpha (unambiguous plaid).

And implement the staircase procedure on every trial with 3 steps up and one step down to at least get an 80% success with the time of each trial to 30 seconds.

The schematic of this experiment is shown in figure 9. Figure 9a is unambiguous plaids without dots with ambiguous plaid in the upward direction. Figure 9b is ambiguous plaids with dots and Figure 9c is unambiguous plaids with dots with ambiguous plaid in the upward direction.

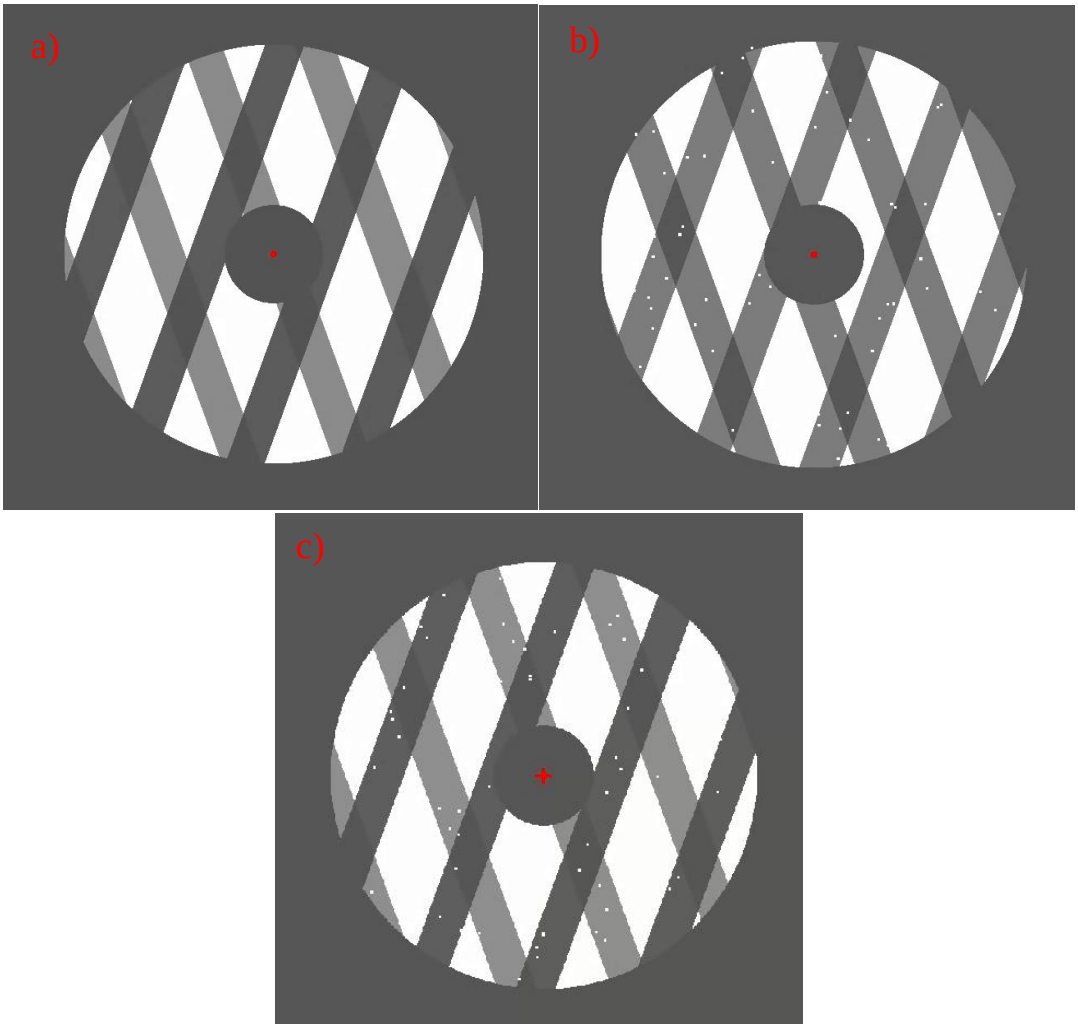

Figure 9 – First design of the experiment, a) unambiguous plaids without dots with ambiguous plaid in the upward direction. The alpha value of the gratings in this figure is 0.85 for the left gratings and 0.8 for the right gratings b) ambiguous plaids with dots, c) unambiguous plaids with dots with ambiguous plaid in the upward direction.

3.4. Subjective Test

due to the lack of time on the project and wanted to get at least some data with the real subject, as we discussed with Professor Phlypo, I changed the design of the experiment and the final design was just include the ambiguous plaids with dots and with constant alpha. and for every direction of left, right and, up, I put a 3 level of coherence for the dots (10%,50%,90%). At the beginning of the experiment, we put a baseline aperture with zero percent coherence on the dots. and due to the previous study, that most of the time human mind has an upward percept on the direction, we instead use left and right direction. The time of this base experiment was longer (duration was 15s) to adapt the participant to the experiment and also when the mind understands about the experiment, it needs less time for the further trial. Then 9 apertures were shown to the participant with the first 3 with supposedly left direction

and 3 levels of noise, then the other 3 with up direction and the last 3 with the right direction. Every stimulus only played for 3 seconds and then turned off and was asked the participant to choose the direction. These recorded data were then analyzed statistically instead of using psychophysical adaptation (staircase).

4. Results

The procedure aims to collect an absolute subjective opinion on the direction of each stimulus. In this way, we develop a program for psychophysics experiments with the psychopy library of Python. Before the start of the experiment, a training session as a baseline to the perception of each viewer was designed with 15 seconds duration on transparent plaids containing the dots with zero percent coherence. These results are not taken into account in the final data.

For the final experiment due to the lack of time, a video of the experiment was generated and with the description of the experiment handed to the participant and ask them to give their percept on every stimulus.

There were 10 participants in this experiment 5 females and 5 males with different visual conditions. Also, it is not neglectable that because the experiment didn't do in the same place and other constant factors such as light, distance to the screen, the size of the screen, and also the time they experimented, I don't have enough confidence in the results. The result has taken without a supervision of a person familiar with the experiment and some participants had a problem in the middle of the experiment and tried it twice from the beginning.

Some of the participants had a hard time deciding on the left and right direction of the plaids on the transparent motion but when I force them to choose between the right and the left, 3 of them did the experiment another time and decided and change their perception from transparent to right or left but other 2 participants told me later that they randomly decided on the direction in the case of transparent motion. I had 2 participants among 10 that only could see the plaids in coherent motion (upward direction) on all of the stimuli.

The final acquired data analyzed in Python and R-studio to be shown and analyze statistically.

The Left, Up, and, Right directions were assigned 1,0, -1 in the analysis of the data. These directions define as the direction of the dots and the correct direction of the plaids that are based on the initial hypothesis supposed to be taken from the subject percept.

figure 10 shown the perception of the subjects when the direction of the dots was tended to left and the data shown in the figure categorized based on the level of noise in the dots.

figure 11 shown the perception of the subjects when the direction of the dots was tended to up and the data shown in the figure categorized based on the level of noise in the dots.

figure 12 shown the perception of the subjects when the direction of the dots was tended to right and the data shown in the figure categorized based on the level of noise in the dots.

Figure 13 represents the mean and standard deviation (sd) of each direction based on the 3 levels of noise presented above.

The right and wrong answer shown in figure 13 was meant that the result we expected to get but the subject perceives otherwise in the direction of the plaids.

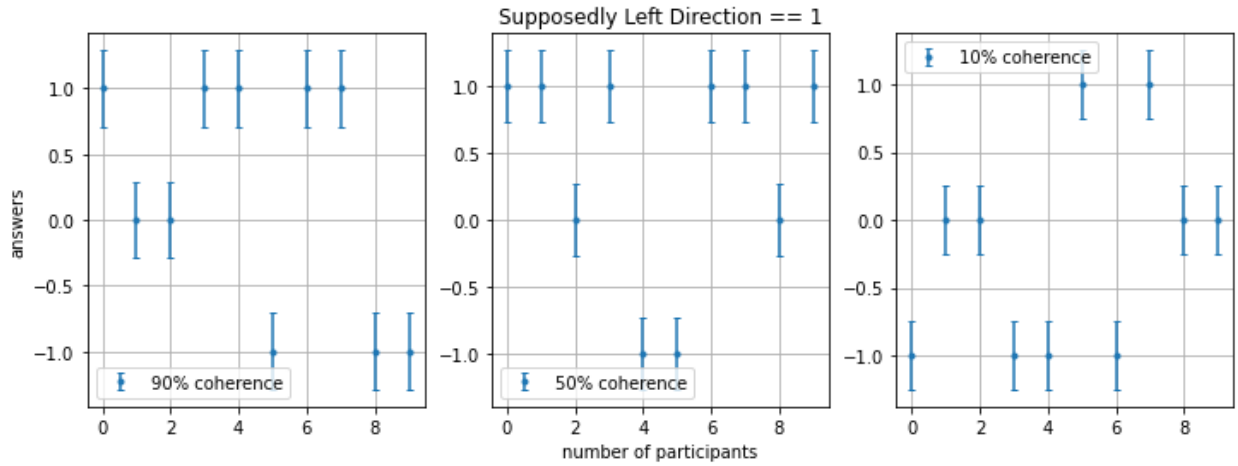

Figure 10 - the perception of the subjects when the direction of the dots was tended to left and the data shown in the figure categorized based on the level of noise in the dots

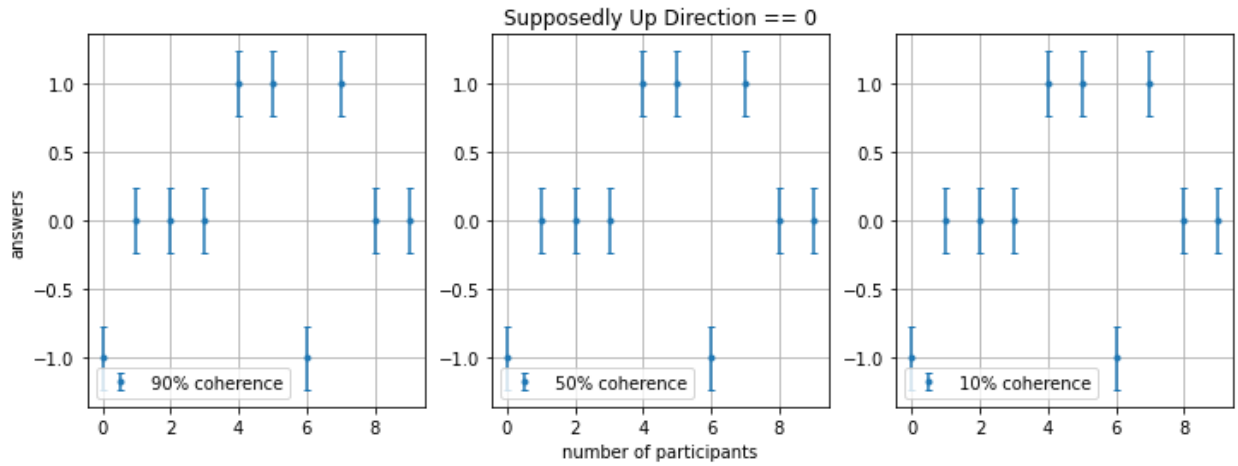

Figure 11 - the perception of the subjects when the direction of the dots was tended to up and the data shown in the figure categorized based on the level of noise in the dots

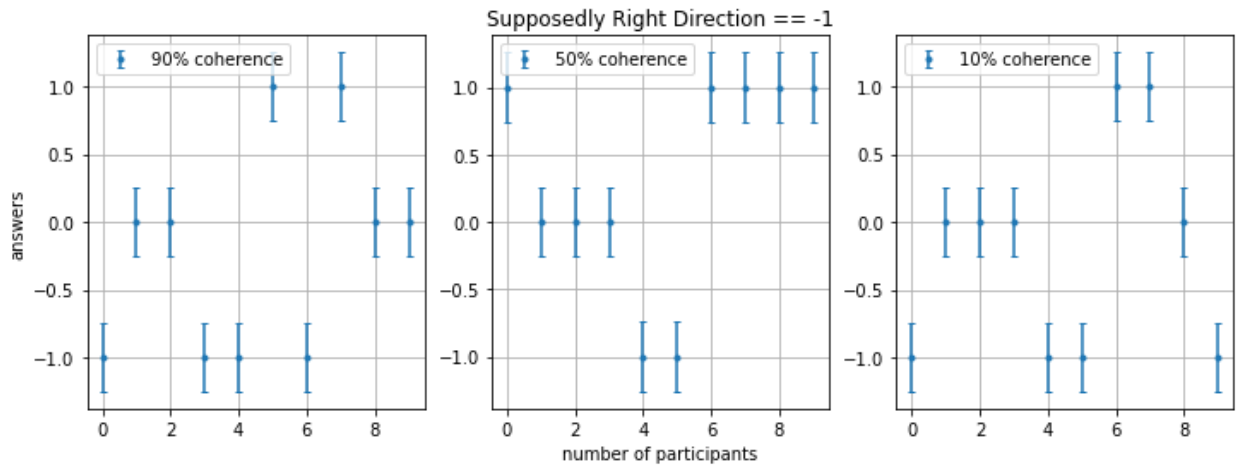

Figure 12 - the perception of the subjects when the direction of the dots was tended to right and the data shown in the figure categorized based on the level of noise in the dots

All of the subject's perceptions are gathered in Table 1. The first column is the dots and expected direction of the plaids. Column 2 to 11 is the subject's answer to the 9 stimuli. The last column is the coherence level of each stimulus. The data of the base experiment is excluded from this table.

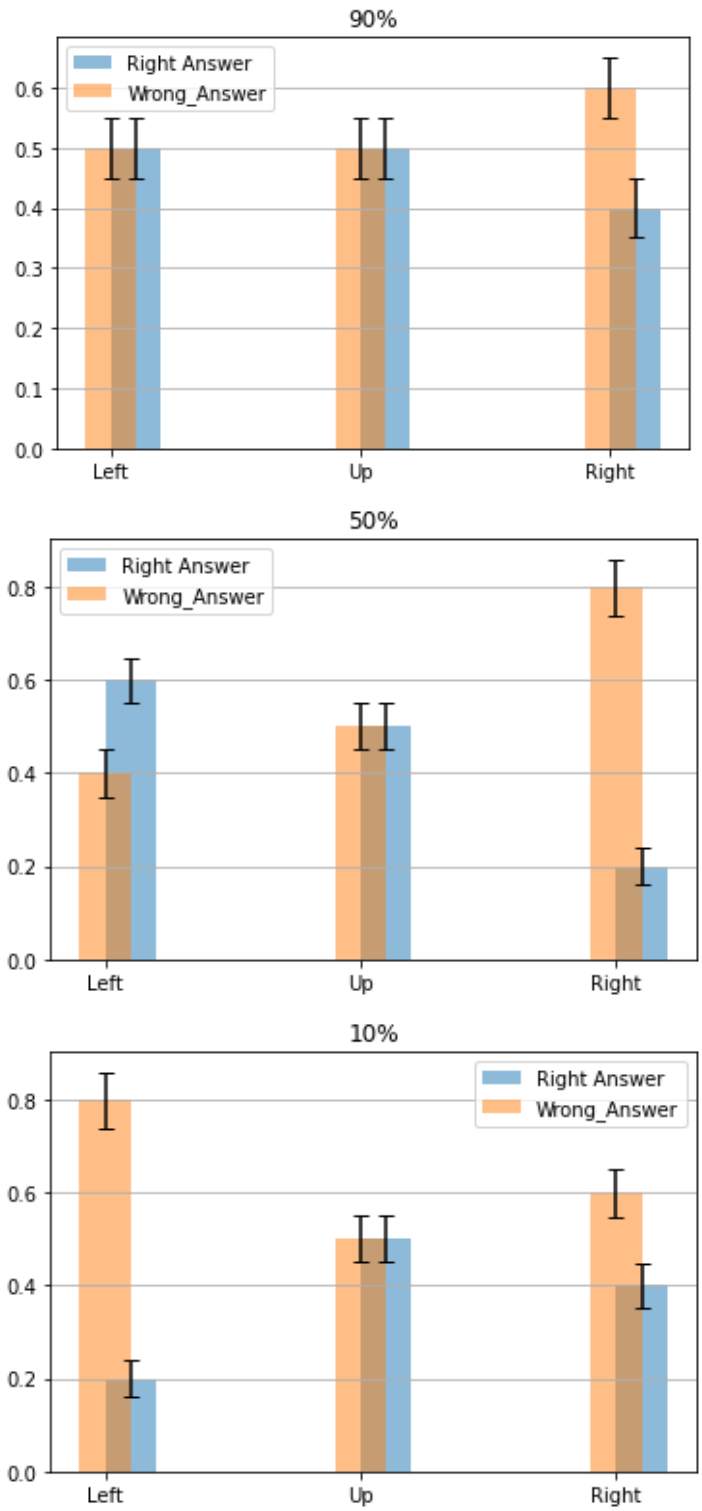

Figure 13 - the mean and standard deviation (sd) of each direction based on the 3 levels of noise. The right and wrong answer that shown in the figure was meant that the result we expected to get but the subject perceives otherwise in the direction of the plaids.

Table 1 - The first column is the dots and expected direction of the plaids. Column 2 to 11 is the subject's answer to the 9 stimuli. The last column is the coherence level of each stimulus.

CorrAns	Part1	Part2	Part3	Part4	Part5	Part6	Part7	Part8	Part9	Part10	Coherence
left	left	up	up	left	left	right	left	left	right	right	0.9
left	left	left	up	left	right	right	left	left	up	left	0.5
left	right	up	up	right	right	left	right	left	up	up	0.1
up	right	up	up	up	left	left	right	left	up	up	0.9
up	right	up	up	up	left	left	right	left	up	up	0.5
up	right	up	up	up	left	left	right	left	up	up	0.1
right	right	up	up	right	right	left	right	left	up	up	0.9
right	left	up	up	up	right	right	left	left	left	left	0.5
right	right	up	up	up	right	right	left	left	up	right	0.1

5. Analysis

To test the effect of noise and direction of random dots kinematogram on the human percept on the direction of an ambiguous plaid, the analysis of variance (ANOVA) was calculated in the R-studio based on the subject's data.

To test for statistical differences, I computed a repeated-measures ANOVA on perception with coherence level and directions as factors. I obtained no significant result on coherence level of 90 and 10 percent ($F(1,3) = 0.698$ $P=0.4869$, $F(3,9) = 0.658$, $P=0.5121$), but there was a significant result on 50 percent coherence ($F(2,6)= 1.810$ $P=0.0737$), and no significant main effect of direction ($F(2,8) = 0.233$, $P = 0.995$). The interaction between coherence and direction was also not significant ($F(6,24) = 1.810$, $P = 0.251$). This indicates that there was no general relationship between coherence and the direction of the dots with the human percepts on the general direction of the plaids. The effect of the 50% coherence is not acceptable due to the condition of the experiment.

The result of this experiment showed no significant effect on the manipulation of a human percept on the direction but there is still the possibility of corrupted data and the small number of trials and short period that causes no significant effect. So, it is difficult to conclude to reject this hypothesis. it has the prospect of improving and the initial idea stands by itself and based on these results it can't be completely rejected.

6. Conclusion

One of the best useful tools for psychophysicists to investigate visual motion detection is Random Dot Kinematograms. In this project, we study the effect of RDKs and the bistable moving paradigm one among the mechanisms controlling visual perceptual bistability. We wanted to show that by putting RDKs on the plaids of the moving paradigm and by changing two factors on the dots (noise and direction) we may obtain to change a human percept to perceive the direction of the ambiguous plaids according to the direction of the dots. we tried to implement and improve some methods on creating the plaids and designing the aperture as the dots only appear on the plaids and also improved the velocity of the dots and the plaids to be correlated to each other and it changed by time and some factor of the whole aperture.

After some objective tests and some try-and-error sessions instead of doing a psychophysical staircase procedure, we switched to the statistical analysis.

Then the obtained data analyzed with ANOVA methods, however, no significant effect was obtained from noise and direction of the dots on human visual percept of the direction of the plaids.

Also, there was no significant effect on the noise and the direction, there was still an effect based on the result of this experiment. If instead of defining the direction of the plaids in three different directions, defining them on coherence and transparent direction, based on the previous study that most of the time the subject perceives the direction as coherence, this study proves otherwise. Most of the subjects in this experiment expressed that they saw transparent motion and sometimes it was hard for them to choose between the direction but it was on the transparent motion on the period of the trial.

This matter and some other factors may have a better effect such as the number of the dots on the plaids or putting the dots in the periphery instead of the center of the aperture.

These are the matters that need further study and need more research and discussion.

Appendix A

Date and Time:			
Civility:			
First name:	Last name:	Age:	Profession:

Do you wear glasses?
How much time have you been looking at electronic devices today?
How much time have you been looking at books today?
On scale of 1 to 10, How do you explain you energy level? (1 is exhausted -- 10 is energetic)
Do you have any other problems regarding your eyes?
Do you have problem in concentrating?
Do you use drugs?

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